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Rewilding psychology

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Some commentators have recently argued that scientific psychology is overly reliant on artificial laboratory-based activities and that it undervalues field-based investigations. However, it remains unclear how a field-based programme of psychological research might be organized in a scalable way. We examine and compare two existing field-based approaches: Roger Barker's behaviour settings programme and Edwin Hutchins's distributed cognition programme. Both programmes prioritize observational work, and both reject the individual as the unit of analysis in favour of a community-scale unit. However, whereas the behaviour settings programme is concerned with structural properties of community life, distributed cognition is concerned more narrowly with the functional analysis of expert team performance. We discuss how these programmes can inform a future community-scale approach to studying psychology in the wild. We conclude that the two programmes are proof of concept of the possibility of a scientific psychology that rejects methodological individualism.

This article is part of the theme issue 'People, places, things and communities: expanding behaviour settings theory in the twenty-first century'.

1. Introduction

In recent years, the field of psychology has been experiencing a prolonged period of crisis [1]. In the early 2010s, researchers began to worry about the non-replicability of well-known findings [2]. Psychology's replication crisis was born [3]. Separately, but around the same time, researchers also came to recognize the problem of the non-representativeness of participant samples used in psychology experiments. Most participants in psychology experiments are from WEIRD backgrounds—they come from Western, Educated, Industrialized, Rich and Democratic (WEIRD) societies [4–6]. More recently, attention has turned to other problems for experimental psychology, including the non-generalizability of findings from empirical studies [7] and the overly 'domesticated,' or laboratory-bound, nature of much psychological research [8,9].

In fact, the roots of these various crises are not new. For as long as psychology has existed as a distinct discipline, psychologists have repeatedly voiced criticisms of the field's direction of travel. One important issue has been the central role of laboratory-based studies in psychological investigation. Experimental psychology has traditionally been a laboratory-based discipline, drawing its facts from studies in which individual participants are asked to engage in artificial tasks [10]. However, psychologists are ultimately also interested in understanding how minds work in their natural setting, including in interaction with other people [11]. Psychology would seem to be in need of a broader set of methodologies that go beyond artificial laboratory-based tasks performed by individuals. Indeed, some researchers have, over the past decades, attempted to establish field-based programmes for

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conducting psychological research in the wild. In this article, we focus on two of these field-based programmes. We examine, first, Roger Barker's behaviour settings programme (1947–1972), and, second, the distributed cognition programme developed by cognitive scientists primarily working in southern California in the 1980s and 1990s—here, we specifically focus on the work of Edwin Hutchins, as presented in his book *Cognition in the wild* [12]. Both of these programmes, in different ways and different historical contexts, were set up as systematic attempts to overcome the narrowness of laboratory-bound psychology. Both programmes can be understood, in other words, as attempts to rewild the study of the mind.

To measure how successfully these two programmes overcame the constraints of laboratory-based psychology, we use an analysis framework developed by Sanches de Oliveira & Baggs [6]. This framework allows us to evaluate the scope of the research conducted within these two programmes along four dimensions of analysis: (i) the participants involved in the empirical work, (ii) the background theoretical assumptions, (iii) the methodological tools employed, and (iv) the infrastructure supporting the research. We end with a discussion of the prospects for a renewed empirical programme for studying cognition in the wild, including in community settings. However, first it will be useful to consider how we got here. How did scientific psychology come to be 'domesticated' [8]? In other words, how did scientific psychology come to be bound up with institutional settings such as the laboratory or the classroom?

2. How psychology became domesticated

The beginnings of psychology as a scientific discipline are conventionally traced back to 1879 when Wilhelm Wundt established the first psychological laboratory at Leipzig [10,13]. Prior to this event, psychological topics had been investigated by an assortment of scholars working across different fields, including philosophers, physiologists and theologians. The establishment of the psychological laboratory as a specialized space created a boundary that demarcated a new discipline [10]. From this point on, psychology—or, at least, experimental psychology—would be understood as the science concerned with the systematic measurement of human mental life and behaviour, with measurements being obtained through specialist instruments and methods appropriate to the laboratory setting.

This vision of psychology as paradigmatically a laboratory-based discipline has remained dominant ever since. Not surprisingly, the most significant achievements of psychologists have been obtained through laboratory-based methods. There have, however, long been critics of this narrow understanding of the field. Some have argued that laboratory-based experimentation is inadequate for understanding human life in all its diversity. One of these critics was Wundt himself, who held that, in addition to laboratory-based scientific psychology, a second discipline was needed, namely a 'cultural psychology,' which would investigate historical processes including language and cultural differences [14].

While laboratory-based experimental psychology has thrived as a discipline since Wundt's time, it is arguably the case that the second strand of psychology which Wundt envisaged, cultural psychology, has yet to be fully realized [15]. In fact, the success of the first strand of psychology has arguably created an obstacle to the development of the second strand. Specifically, the success of laboratory-based psychology may have created a narrow set of expectations about what scientific psychology can, in principle, look like: reliant on quantification and statistical methods, performed in a controlled environment, and so on. This way of doing psychology has produced over a century of useful research, but it is research of a kind that builds fundamentally on a number of assumptions about minds and behaviour—assumptions that arose specifically from the WEIRD cultural setting in which psychology developed [10,16]. As a result, experimental psychology as it has come to exist today is fundamentally WEIRD in a number of different ways which we will unpack below [6].

First, it will be useful to summarize what we might call the classic WEIRD problem or the problem of WEIRD participants [6]. The WEIRD acronym was first introduced in Henrich and colleagues' paper, '*The weirdest people in the world*' [4]. The focus of the authors was on the non-representativeness of the participants in psychology experiments. Henrich *et al.* argued, firstly, that experimental psychology studies are overly reliant on college student participants, particularly students living in the United States (US) and western Europe. Secondly, they argued that this sampling bias is problematic because college students are psychological outliers when compared with the broader adult population. In fact, neither of Henrich's *et al.*'s observations were novel [17]. As early as 1946, one researcher wrote: 'The existing science of human behaviour is largely the science of the behaviour of [college] sophomores. Too much research effort is expended on college students with the subsequent waste of journal space devoted to speculation concerning whether the findings hold for mankind in general' [18, p. 333]. In 1966, another researcher noted that 'the college population differs markedly from the adult non-college population in many characteristics which are associated with psychological factors, e.g. age, social class, learning ability' [19, p. 115]. Henrich *et al.*'s important contribution was twofold: first, they provided the field with a memorable label for these problems—college students are WEIRD—and second, they provided a thorough and systematic examination of the scale of the problem.

Henrich *et al.*'s intervention was timely, arriving at around the same time as the replication crisis began. The introduction of the WEIRD label has had a salutary effect on the field in terms of raising awareness of the problem of the non-representativeness of participant samples. More than a decade later, however, there has been little actual change in terms of the participant samples that are used in published research reports [20]. Recently, Henrich and co-authors have complained that 'the needle hasn't moved' in the years since their original intervention [1].

Why has there been little change in the participant samples that psychologists make use of? An obvious reason is that college students remain the most conveniently available population. However, there are deeper considerations that go into the designing and carrying out of empirical psychological work. Making psychology more diverse cannot be achieved merely by sampling from a larger population. The reader of the original '*Weirdest people in the world*' article [4] may be led to conclude that simply by recruiting participants from a more diverse pool, researchers will be justified in drawing conclusions about

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Table 1. Four aspects of the WEIRD crisis in psychology, based on Sanches de Oliveira & Baggs [6].

participants	disproportionately drawn from populations in Western, Educated, Industrialized, Rich and Democratic (WEIRD) societies; predominantly university students in North America and Europe
theory	theoretical assumptions that inform research design. In particular: – individualism: explanation for behaviour is sought within individual minds or bodies – universalism: 'human nature' and individuals as a window to it
methods	tools, tasks and procedures used to generate data: – artificial settings: the laboratory, online surveys – artificial tasks: questionnaires, schoolwork-like activities, logic puzzles, etc. – artificial measures: e.g. reaction times, intelligence quotient
infrastructure	the institutional machinery that enables research, including: – publishing opportunities – funding opportunities – diversity among researchers – access to international research networks, meetings, etc.

humanity as a whole [21]. However, this overlooks the difficulties involved in studying diverse populations. When researchers design an experiment involving students in the US, the researchers necessarily make certain theoretical assumptions and methodological choices. It is often not the case that these assumptions and choices can straightforwardly be transferred to the study of a non-WEIRD population. Henrich [22] gives an example from a study of a population of Hadza in Tanzania who behave differently than US participants when asked to answer questions designed to elicit the endowment effect, a well-known effect in behavioural economics. The interpretation of the responses that these participants gave is complicated by the fact that Hadza customs of ownership are different from those of the Western researchers who originally formulated the endowment effect—so different that, rather than concluding that the Hadza do not display the endowment effect, it is reasonable to wonder whether the category even applies to them. It is not clear that it is, in fact, coherent to talk of an endowment effect with respect to the Hadza population.

The classic WEIRD problem, then, is concerned with the overly narrow participant pool used in psychological studies. Recent work by Sanches de Oliveira and Baggs proposes that the WEIRD crisis in psychology is, however, much broader than the question of how to recruit a more representative participant sample. Representativeness is only the first aspect of the WEIRD crisis in psychology. This first aspect we refer to as the problem of WEIRD participants. Psychologists interested in the cross-cultural study of human mental life must face up to this problem of who the participants are in psychology experiments, but these researchers must also address questions around psychology's deep theoretical assumptions, its methodological choices, and its institutional infrastructure which supports the generation and dissemination of research. These issues are summarized in table 1. A more detailed discussion can be found in Sanches de Oliveira & Baggs [6], which we will briefly summarize here.

The second aspect of the WEIRD crisis in psychology is the problem of WEIRD theoretical assumptions. Two assumptions stand out, namely individualism and universalism. Individualism refers to the assumption that behaviour is to be explained by reference to processes at the scale of the individual; for instance, action is assumed to be explainable as the output of an individual's internal mental processes [23]. This assumption is foundational to laboratory-based psychology [10,16].[1] However, the assumption falls apart upon contact with the real world. For instance, it fails to account for the fact that children are able to successfully perform some tasks with the guidance of a carer, even though they fail to complete the same task when left to solve it by themselves—educators know this as the zone of proximal development [26].

Universalism, meanwhile, refers to the assumption that it is possible to generalize over the set of all human individuals, in terms of psychological traits, because as humans we share a single universal essence. In the words of Montaigne, this is the view that 'every man beareth the whole stampe of humane condition' [27, p. 179], or in more contemporary terms it is a 'view of human nature as variations on a species-typical theme' [28, p. 626]. There is certainly something attractive about this assumption: it is a view of humanity that lends itself naturally to a belief in democracy. However, universalism does not always lend itself to the generation of empirically well-formulated research questions. See the Hadza example above. The assumption of universalism sometimes leads behavioural economists to show up among isolated tribes and to ask the people there to perform a set of economic games that were developed by researchers involved in Cold War strategizing about decision-making under conditions of uncertainty. The question of whether the population being tested has a concept of ownership, or even a concept of money, is considered only as an afterthought. These two assumptions, individualism and universalism, are themselves contingent products of the way that psychology has been developed in the global West, but they can easily be 'mistaken for universal principles having eternal validity' [16, p. 22].

[1]Experimental psychology is traditionally committed to methodological individualism, i.e. to the assumption that explanations for behaviour should refer only to factors internal to the individual actor [24]. The cognitive revolution led to a stronger version of this assumption, according to which explanations for behaviour should refer only to symbolic content, and processes for manipulating that content, that are posited to exist in the mind of the individual. This latter assumption was explicitly defended by Jerry Fodor under the rubric of methodological solipsism [25]. Cognitivism's rise in the second half of the twentieth century thus tended to strengthen psychology's pre-existing bias in favour individualistic explanation.

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The third aspect of the WEIRD crisis is the problem of WEIRD methods. Psychological experiments are often carried out in institutional, or domesticated, settings—the laboratory, the classroom and the clinician's office. This is convenient for experimenters but means that experimenters are only sampling a limited and highly artificial slice of human behaviour. The tasks themselves often bear little resemblance to the kinds of meaningful activities that people do outside the laboratory, leading some to worry that psychology is really 'the science of self-reports and finger movements' [29].[2] The concept of an intelligence quotient (IQ) is an illustrative case. This is often understood as a universal property of humans, one in which individuals can differ only in magnitude, yet it is measured by the use of tasks of a sort that were originally developed to test French school children in the early twentieth century: IQ might better be understood as a measure of aptitude for completing schoolwork [14].

Fourth is the problem of WEIRD infrastructure. This is arguably the biggest problem of all. In order to overcome the Western-centric bias of experimental psychology, it would seem necessary that non-Western researchers be involved in the identification of research questions, the development of methodologies, the allocation of research resources and the dissemination of findings [31,32]. At present, substantial inequalities exist in all of these areas [20,33,34].

Our purpose in this article is to evaluate two research programmes that have attempted to expand the scope of psychological research beyond the laboratory and into the real world. Roger Barker's behaviour settings programme and Edwin Hutchins's ethnographic work in the distributed cognition tradition were two attempts to rewild psychology. In the following two sections, we summarize these two programmes in terms of their respective negative critique of laboratory-bound psychology and their positive programme for studying behaviour in natural settings. Both programmes can, retrospectively, be understood as attempts to pursue a non-WEIRD approach to psychology. We thus evaluate the two programmes using the framework set out above: who are the participants studied? What theoretical assumptions are made? What methods are used? What infrastructure is necessary to support the research programme?

3. Roger Barker's behaviour settings programme

(a) Overview

Roger Barker's behaviour settings programme was a uniquely ambitious project in twentieth-century psychology. Barker and his colleagues sought to record the public behaviour patterns of an entire community over a number of years. The main operating hub was in the small Kansas town of Oskaloosa, codenamed 'Midwest' in publications, where Barker and his colleagues lived, and where Barker ran a psychological field station between 1947 and his retirement in 1972. Barker also ran a second operation in the English town of Leyburn, North Yorkshire, codenamed 'Yoredale' [35]. The major contributions of the programme are the concept of behaviour settings and the hypothesis that underpopulated settings elicit more diverse behaviour from the inhabitants when compared with optimally populated or overpopulated settings.

(b) Barker's critique of experimental psychology

Barker's programme is motivated by a sharp critique of experimental and social psychology as it existed in the mid-twentieth century. Barker's insight was that psychologists, in sampling behaviour only in the laboratory, were ignorant about the distribution of behaviour in the real world. This critique is clearly stated in an application for funding Barker wrote in 1947:

> Although we know a great deal about how children behave under relatively controlled, standardized conditions, such as intelligence- and personality-testing situations, we know little about the nature of the situations that actually confront children in their daily lives and how they react to them. Preoccupation by investigators with behaviour under controlled conditions has crowded out concern with naturally occurring situations and with the interrelations between them and behaviour. We do not know in concrete detail how parents, teachers, and other adults and children behave towards children, we do not know what pressures and demands are made upon children, we do not know how they respond: what successes, failures, frustrations, and happinesses children actually experience. [36, p. xi].

Barker often returned to the project of sharpening and elaborating this critique. Barker's major breakthroughs, however, came through critically engaging with his own ongoing work.

(C) Major contributions

The signature conceptual innovation of Barker's research programme is the concept of behaviour settings. A behaviour setting is a repeating pattern of behaviour that exists at a scale larger than the individual actor and includes both collectives of actors and the objects and locations in which the action takes place. Examples include a church service, a high school play or a post office during operating hours [37]. Early on in their investigations, Barker and his colleagues attempted to sample the behaviour of children in Oskaloosa by following individual children around and recording the behaviour of these individual children

[2]The fact that a study uses a highly artificial procedure does not necessarily imply that the study serves no useful purpose. The aim of some experimental studies is not to generate new facts about behaviour but to test whether existing generalizations about behaviour are correct. Mook [30] cited the example of Harry Harlow's famous cloth versus wire monkey study, which used a deliberately artificial procedure to produce results which were nevertheless inconvenient for then-popular drive-reduction theories in psychology. In Mook's analysis [30], the purpose of artificial tasks is not to recreate reality outside the laboratory but to test theories about reality outside the laboratory.

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Table 2. Summary of Barker's behaviour settings programme in terms of the categories outlined in table 1.

participants	recruit entire community, small towns in Kansas and North Yorkshire, high schools, 'behavioural research without subjects' [42]
theory	non-individualistic; universalist in the sense that behaviour settings are held to be natural units; units of analysis are behaviour settings; structure-driven
methods	ethnographic methods; compiling pre-existing data sources, e.g. newspaper notices
infrastructure	Midwest psychological field station; work published largely in monographs and edited volumes

in minute detail [38]. By developing the concept of a behaviour setting, the researchers were able to switch from the scale of individual behaviour to the scale of group activity. This proved fruitful. Barker wrote that 'we could predict some aspects of children's behaviour more adequately from knowledge of the behaviour characteristics of the drugstores, arithmetic classes, and basketball games they inhabited than from knowledge of the behaviour tendencies of particular children' [39, p. 4].

Having developed the concept of a behaviour setting, Barker and his team sought to use it to intervene in policy debates, one of which concerned the consolidation of public high schools into larger schools. Barker *et al.* conducted a major investigation comparing differently sized high schools in Kansas [40]. Through this work, Barker developed his second major contribution: a theory of underpopulated settings. According to the theory, pupils in small schools tend to participate in a greater diversity of settings, when compared with their big school counterparts. Barker and his co-author illustrated the theory by describing a junior class play. In a small school, 'each member of the class is essential to the play's successful presentation, versus the Junior Class play of a large school, where at most only 15 per cent of the members of the class have more than a spectator's involvement in the play' [40, p. 195]. Based on their observations, the authors argued that 'A school should be small enough that students are not redundant' [40, p. 202].[3]

(d) Discussion

Barker's behaviour settings programme was motivated by some of the same considerations that have fed into the more recent narratives of crisis in experimental psychology discussed above. We will now briefly discuss Barker's programme in terms of our four categories of analysis: participants, theory, methods and infrastructure (summarized in table 2).

By setting up a field station in a small town, Barker deliberately rejected the model of conducting psychological studies that relied on undergraduate participants. Barker's ambition was to recruit an entire community, starting with the school children. As the research programme went on, however, Barker *et al.*'s focus shifted away from individual participants altogether. They began to focus instead on the behaviour settings themselves as their primary units of analysis. After his retirement, Barker described his research project as 'behavioural research without subjects' [42, p. 30].

As a result of this shift in emphasis, Barker's programme is hard to place with respect to classical laboratory-based psychology. The behaviour-setting concept offers a non-individualist lens on human behaviour but at the cost of relegating individual minds themselves to the background [41]. For this reason, Barker's programme is ill-suited to generating universalist discoveries at the scale of a hypothetical human nature. Nevertheless, Barker's programme does seek to make universalist discoveries, as with the theory of underpopulated settings which leads to the assertion that schools, as a general rule, should be small in size. This is a universalist statement about the architectural properties of behaviour settings rather than about psychology *per se*.[4] Over time, Barker and colleagues stopped describing what they were doing as 'ecological psychology' and instead favoured the label 'ecobehavioural science' [43].

Consistent with the shift in theoretical emphasis, the Barker programme also underwent a shift in methods. Whereas the early work involved ethnographic observation of individual behaviour [38], the later work relied heavily on pre-compiled data sources about the behaviour settings that existed in a given milieu, such as local newspaper notices, school publications and municipal statistics [36,39,40].

Barker's research programme is unique in terms of the infrastructure involved in supporting it. Most important was the town of Oskaloosa itself. The town was situated far enough away from neighbouring settlements that much of its activity was self-contained. Not all towns in the US in the 1940s had this property. Before he was recruited by the University of Kansas, Barker had earlier looked into the possibility of running a similar field study in Massachusetts. However, he found the towns there uncongenial:

> The small settlements of the region were not self-contained towns as in the Midwest; they were spacially [*sic*] dispersed and often specialized fragments of political units (towns) often straggling many miles along streams or across forested ridges. The children of these fragments did not have a common community environment as did the children of the towns I had seen in Illinois. [44, p. 2152]

This anecdote in itself draws into question the universality of the behaviour settings programme. To what extent is it possible to generalize from the settings of Oskaloosa to the settings of other towns in the US, let alone to settings of places elsewhere in the world? We will discuss the scope and limitations of the behaviour settings programme below.

[3]For a critical discussion of the theory of underpopulated settings, see Perkins *et al.* [41].

[4]The Barker programme also sought to develop tools for identifying and individuating, in a rigorous and generalizable manner, the different behaviour settings that exist in a community. This led the researchers to develop the K-21 test, which Barker *et al.* outline in detail in their published work (e.g. [39]).

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Another piece of infrastructure that is necessary for the continuation of any research programme is the possibility of disseminating results. Barker's own writings were largely published in his monographs and in his edited volumes in which he presented his own chapters alongside work by his colleagues. The books still offer a rich set of resources to the reader today. Many of these texts, however, have long been out of print, making it difficult for potentially interested researchers to discover his work. No international society or journal has ever been set up exclusively for the dissemination of ecobehavioural science, and as a result, the programme ran out of energy in the years after Barker's retirement, though Barker continued to hope for a revival [42].

4. Edwin Hutchins and the distributed cognition programme

(a) Overview

Whereas Barker's programme was a response to the behaviouristic psychology that was ascendant in the 1940s, the distributed cognition programme was a response to the cognitive science of the 1970s and 1980s. The distributed cognition programme builds on the insight that cognitive work is carried out not only inside people's heads but also in the manipulation of symbols and objects in the environment. The framework has lent itself particularly well to applications in the fields of human–computer interaction and design [45,46]. We focus here on Edwin Hutchins's work with expert teams. Like Barker, Hutchins chose to investigate human behaviour at the scale of the group. However, where Barker's team focused on describing the structural organization of behaviour settings in a community, Hutchins's work focuses on the functional organization of group behaviour with respect to a shared activity. In his book *Cognition in the wild*, Hutchins [12] offers an insightful analysis of maritime navigation practices in two expert group settings: on board a US Navy vessel and on canoes used in traditional Micronesian navigation.

(b) Hutchins's critique of cognitive science

Cognitive science is built on the metaphor of the mind as a machine [47]. In the classical formulation of the computer metaphor, the brain is understood as a physical symbol system that operates by performing computations over abstract symbols [48]. In the final chapter of *Cognition in the wild*, Hutchins offers a pointed critique of this metaphor. Hutchins argues that, for humans, symbols are not innate properties of our brains, but are first encountered as external artefacts used in social situations, such as the navigation charts and instruments used on board the aircraft carrier. The physical symbol system hypothesis abstracts away from both the materiality and the social nature of people interacting with actual objects. For Hutchins, this means that artificial intelligence research is flawed because it equates thinking with the symbols themselves while abstracting away from the social activities that give meaning to those symbols:

> Having failed to notice that the central metaphor of the physical-symbol-system hypothesis captured the properties of a socio-cultural system rather than those of an individual mind, AI and information-processing psychology proposed some radical conceptual surgery for the modelled human. The brain was removed and replaced with a computer. The surgery was a success. However, there was an apparently unintended side effect: the hands, the eyes, the ears, the nose, the mouth and the emotions all fell away when the brain was replaced by a computer. The computer was not made in the image of the person. The computer was made using images of formal manipulations of abstract symbols. And the last 30 years of cognitive science can be seen as attempts to remake the person in the image of the computer. [12, p. 363]

As this passage makes clear, Hutchins was concerned with reinstating the experiencing human agent to explain the performance of cognitive tasks, such as staying oriented to the maritime environment.

(c) Major contributions

A major contribution of the distributed cognition framework is that it allows for an analysis of cognitive performance across teams involving multiple people. Hutchins's ethnographic work in his book is based on video and audio recordings and notes taken during observation of a navigation team on board a US Navy aircraft carrier [12, p. 22]. Hutchins's video captures an overhead shot of the naval chart, while an audio track captures the verbal communication between team members over a telephone circuit. This set-up allows Hutchins to analyse the collective work performed to keep track of the position of the ship. The task of computing the position of the ship during the 'fix cycle' involves up to six people, each performing sub-components of the overall task, such as looking through telescopes or noting down the bearings of landmarks [12]. Another important theoretical contribution of Hutchins's work is the concept of a material anchor, i.e. a physical component of the environment that plays a direct role in cognitive work [49]. In a separate analysis of navigation on board an aircraft, Hutchins [50] describes how the airspeed indicator dial is an important component in the task of coordinating the speed of the airliner as it approaches the runway during landing. The dial has an adjustable marker, which is set by the pilots into a fixed position ahead of the landing descent in order to indicate the desired reference speed for the aircraft as it lands. Once this marker is set in place, the pilots can monitor whether the plane is travelling in the desired speed range simply by looking to see whether the speed indicator needle is lined up with the marker [12, p. 283]. The marker performs part of the cognitive task in that it relieves the pilots of having to remember the relevant numbers: the marker serves as a material anchor for thought. It also serves as a shared, visually accessible resource in the environment.

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Table 3. Summary of Hutchins's distributed cognition programme in terms of the categories outlined in table 1.

participants	recruit expert teams, constrained problem-solving tasks
theory	non-individualistic; universalist at the scale of the computational task; units of analysis are distributed control systems; function-driven
methods	ethnographic methods, video and audio recordings, focus on external symbolic representations and equipment
infrastructure	*SS Palau*, military and industry funding

(d) Discussion

Like Barker's work, Hutchins's work involves participants in the real world. However, where Barker attempts to survey an entire community, Hutchins focuses narrowly on a single expert team working in a highly constrained setting (see summary in table 3). In terms of theory, the distributed cognition framework is non-individualistic. Its units of analysis are distributed control systems—the aeroplane cockpit, including flight crew and instruments; the navigation deck, including charts, instruments and crew; the canoe, including its crew and its entire surrounding visible world, the night sky. Hutchins is able to partly capture the experiences of the individual people within these systems and convey that experience to his readers, by transcribing relevant parts of the participants' speech or by reproducing relevant parts of their visual environment.

As with Barker's programme, there is a universalistic project in Hutchins's work, but not one involving 'human nature.' Instead, the universalism in Hutchins's work derives from the shared functional nature of activity across different cultural contexts. Hutchins [12] adopted the neuroscientist David Marr's distinction [51] between three levels of cognitive analysis—computational, algorithmic and implementational—and applied it to his expert teams. According to Hutchins, the Navy navigation team and the Micronesian navigators are solving the same computational task—both teams are constantly answering the question 'Where am I?' The two teams solve the task, however, by implementing starkly different algorithmic solutions. In terms of methods, Hutchins adopts an ethnographic approach. To document the teams' activities, he makes use of video and audio recordings as well as direct observation. This is demanding work, for it requires the observer to, in some sense, become an expert in the task that they are observing, which can be very technical and complex. The observer must also gain institutional permission to be there and must earn the trust of the people they are observing (a problem that Barker's team also faced). Hutchins's work has been influential, yet, to date there is no journal or international society dedicated to distributed cognition or cognitive ethnography.

5. Scope and limits of the two programmes

As is evident from the discussion above, there are interesting similarities between the two programmes. Both seek to document behaviour as it exists in the real world, instead of in the laboratory; both reject the individual as the unit of analysis in favour of a collective-scale focus; and both programmes are based on a sharp critique of existing behavioural research.

Perhaps the sharpest critique of laboratory-based experimental psychology that Barker published appears in a 1965 paper entitled '*Explorations in ecological psychology*' [52]. He contrasts two models for conducting psychological research, using terminology from cybernetic theory. In the first model, the psychologist acts as a 'transducer,' or in other words, the psychologist goes to the field and simply documents behaviour as it occurs. This is Barker's preferred model. In the second model, the psychologist acts as the 'operator' of the system of behavior under investigation. Laboratory-based experimental psychology requires that the experimenter be the operator of the system in the sense that the experimenter strictly controls the inputs to and the environment of the research participant. A problem with the second model is that it is unclear whether the psychologist is really discovering universal truths about human behaviour, or whether they are instead creating an entirely new pattern of human behaviour as a result of the artificial circumstances that they, the experimenter, have created.

Barker acknowledged, however, that advocates of the first model—psychology as field-based observation—face a difficult task in trying to get their colleagues to pay attention to their work:

> [T]here are a number of reasons for avoiding the role of the transducer in psychological research: The resulting T [transduced] data do not fit easily into the hardware, the bureaucracies, the research designs, or the concepts of psychology's establishment which has been shaped by prevailing operator data-generating systems. The skills and personality attributes required of a successful transducer are different from those of a successful operator, and they are doubtless not congenial to many persons. The techniques of the transducer are, in many respects, more difficult than those of the operator. Most importantly, perhaps, the transducer has less easy access to the organism sector of psychological units, i.e., to psychology's challenging 'black box.' [52, p. 13]

The last sentence here is an acknowledgement that Barker's research, by shifting its focus to impersonal behaviour settings, has effectively given up on the experiential scale of analysis of behaviour—it is an acknowledgement of the switch to 'behavioural research without subjects' [42, p. 30]. Barker's programme is preoccupied with the structural organization of behaviour settings in a community [41]. This is in contrast to Hutchins's work, which deliberately focuses on the experiential scale. Hutchins succeeds in maintaining the experiential scale as a part of the analysis by restricting his investigation to constrained, functional settings where an expert team can be said to have a shared goal. Researchers who are interested in employing

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Hutchins-style cognitive ethnographic methods in less tightly constrained settings face difficulties in defining the scope of the investigation [53].

To summarize, Barker's behaviour settings programme succeeds in sampling behaviour as it occurs in the real world, but it does so at the cost of ignoring experiential processes *per se*. Hutchins's distributed cognition programme maintains a focus on psychological processes but at the cost of restricting its scope of the investigation to a constrained set of behavioural situations involving the manipulation of external symbolic representations.

6. Prospects for a future psychology of community

It would seem customary to end this discussion by suggesting that there are lessons to be learned from the two programmes and that psychologists in the future can build on insights from both. We are hopeful that this is the case.

It would also seem timely to explore the possibility of reviving the study of behaviour at the community scale. Chater & Loewenstein have recently argued that a shift of focus from the individual to the collective is necessary if behavioural scientists are to contribute to the solving of collective action problems such as climate change [54]. As they point out, behavioural economists have traditionally attempted to address societal challenges by influencing the behaviour of individuals, but often a more effective approach is to address the architecture of the entire system. For instance, if a national government wishes to improve the thermal efficiency of the nation's housing stock, the government could publish information about the benefits of insulation and wait for people to pay for installing it themselves (an individualistic intervention), but change will probably occur more quickly if the government mandates and subsidizes installation of insulation (a collective intervention).

A psychology of community may be something that we have a societal need for. We should, however, be aware of the difficulty of the task. The history of psychology suggests that it will not be easy to scale up a community-oriented or cultural approach to psychology in such a way that a large number of psychologists adopt a common set of theoretical tenets and methodological practices [14]. Psychology remains very much attached to the laboratory. As we have seen, the establishment of Wundt's laboratory was the foundational event for psychology as a discipline. Scientific psychology was born as a domesticated animal. Perhaps a wild version of psychology cannot be the same discipline but can only ever be a close cousin.

One important difference between laboratory-based psychology and its field-based counterpart is that, in the case of the former, it is at least in principle possible to replicate someone else's study. A Stroop effect can be elicited from college students anywhere. The success of domesticated methods in psychology can be explained by the fact that they can reliably produce empirical outputs. By contrast, as Hutchins notes, 'Studying cognition in the wild is difficult, and the outcomes are uncertain' [12, p. 371]. The two programmes that we have focused on do, at least, offer a starting point. Barker's programme offers a set of tools for thinking about the structure of activity in a community, while Hutchins's programme offers a complementary set of tools for analysing the functional organization of teams.

If a single lesson is to be learned from the two programmes we have discussed, it would be that it is possible to conduct psychological research while rejecting methodological individualism, or the assumption that the explanation for some behaviour should ultimately be expressed in terms of processes inside the organism. Both the behaviour settings programme and the distributed cognition programme provide proof of concept that it is possible to identify structured units of analysis at larger scales, with individual actors as sub-components in these larger units. Wundt's ambition to pursue a second psychology will perhaps finally be achievable if psychologists can learn to relax their expectation that the individual is the supreme locus of psychological explanation.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.
Data accessibility. This article has no additional data.
Declaration of AI use. We have not used AI-assisted technologies in creating this article.
Authors' contributions. E.B.: conceptualization, writing—original draft, writing—review and editing; G.SdO.: conceptualization, writing—review and editing.

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References

1. Apicella C, Norenzayan A, Henrich J. 2020 Beyond weird: a review of the last decade and a look ahead to the global laboratory of the future. *Evol. Hum. Behav*. **41**, 319–329. (doi:10.1016/j.evolhumbehav.2020.07.015)
2. Simmons JP, Nelson LD, Simonsohn U. 2011 False-positive psychology: undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychol. Sci*. **22**, 1359–1366. (doi:10.1177/0956797611417632)
3. Pashler H, Wagenmakers EJ. 2012 Editors' introduction to the special section on replicability in psychological science: a crisis of confidence? *Perspect. Psychol. Sci*. **7**, 528–530. (doi:10.1177/1745691612465253)
4. Henrich J, Heine SJ, Norenzayan A. 2010 The weirdest people in the world? *Behav. Brain Sci.* **33**, 61–83. (doi:10.1017/S0140525X0999152X)
5. Clancy KBH, Davis JL. 2019 Soylent is people, and weird is white: biological anthropology, whiteness, and the limits of the weird. *Annu. Rev. Anthropol.* **48**, 169–186. (doi:10.1146/annurev-anthro-102218-011133)
6. Sanches de Oliveira G, Baggs E. 2023 *Psychology's WEIRD problems*. Cambridge, UK: Cambridge University Press. (doi:10.1017/9781009303538)
7. Yarkoni T. 2020 The generalizability crisis. *Behav. Brain Sci*. **45**, 1–78. (doi:10.1017/S0140525X20001685)

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8. Ibanez A. 2022 The mind's golden cage and cognition in the wild. *Trends Cogn. Sci.* **26**, 1031–1034. (doi:10.1016/j.tics.2022.07.008)
9. Newson M, Buhrmester M, Xygalatas D, Whitehouse H. 2020 Go wild, not weird. *J. Cogn. Sci. Relig.* **6**, 80–106. (doi:10.1558/jcsr.38413)
10. Danziger K. 1990 *Constructing the subject: historical origins of psychological research*. Cambridge, UK: Cambridge University Press. (doi:10.1017/CBO9780511524059)
11. Dingemanse M *et al*. 2023 Beyond single-mindedness: a figure-ground reversal for the cognitive sciences. *Cogn. Sci*. **47**, e13230. (doi:10.1111/cogs.13230)
12. Hutchins E. 1995 *Cognition in the wild*. Cambridge, MA: MIT Press. (doi:10.7551/mitpress/1881.001.0001)
13. Harper RS. 1950 The first psychological laboratory. *Isis* **41**, 158–161. (doi:10.1086/349141)
14. Cole M. 1996 *Cultural psychology: a once and future discipline*. Cambridge, MA: Harvard University Press.
15. Cahan ED, White SH. 1992 Proposals for a second psychology. *Am. Psychol*. **47**, 224–235. (doi:10.1037/0003-066X.47.2.224)
16. Danziger K. 1997 *Naming the mind: how psychology found its language*. London, UK: Sage. (doi:10.4135/9781446221815)
17. Sears DO. 1986 College sophomores in the laboratory: influences of a narrow data base on social psychology's view of human nature. *J. Pers. Soc. Psychol*. **51**, 515–530. (doi:10.1037/0022-3514.51.3.515)
18. Mcnemar Q. 1946 Opinion-attitude methodology. *Psychol. Bull*. **43**, 289–374. (doi:10.1037/h0060985)
19. Smart RG. 1966 Subject selection bias in psychological research. *Can. Psychol.* **7a**, 115–121. (doi:10.1037/h0083096)
20. Thalmayer AG, Toscanelli C, Arnett JJ. 2021 The neglected 95% revisited: is American psychology becoming less American? *Am. Psychol*. **76**, 116–129. (doi:10.1037/amp0000622)
21. Salari Rad MS, Martingano AJ, Ginges J. 2018 Toward a psychology of *Homo sapiens*: making psychological science more representative of the human population. *Proc. Natl Acad. Sci. USA* **115**, 11401–11405. (doi:10.1073/pnas.1721165115)
22. Henrich J. 2020 *The weirdest people in the world: how the west became psychologically peculiar and particularly prosperous*. London, UK: Penguin UK.
23. Harré R. 1984 *Personal being: a theory for individual psychology*. Cambridge, MA: Harvard University Press.
24. Udehn L. 2002 The changing face of methodological individualism. *Annu. Rev. Sociol.* **28**, 479–507. (doi:10.1146/annurev.soc.28.110601.140938)
25. Fodor JA. 1980 Methodological solipsism considered as a research strategy in cognitive psychology. *Behav. Brain Sci.* **3**, 63–73. (doi:10.1017/S0140525X00001771)
26. Vygotsky LS. 1978 *Mind in society: the development of higher psychological processes*. Cambridge, MA: Harvard University Press.
27. Taylor C. 1989 *Sources of the self: the making of the modern identity*. Cambridge, MA: Harvard University Press.
28. Barrett HC. 2020 Towards a cognitive science of the human: cross-cultural approaches and their urgency. *Trends Cogn. Sci. (Regul. Ed.)* **24**, 626–638. (doi:10.1016/j.tics.2020.05.007)
29. Baumeister RF, Vohs KD, Funder DC. 2007 Psychology as the science of self-reports and finger movements: whatever happened to actual behavior? *Perspect. Psychol. Sci*. **2**, 396–403. (doi:10.1111/j.1745-6916.2007.00051.x)
30. Mook DG. 1983 In defense of external invalidity. *Am. Psychol*. **38**, 379–387. (doi:10.1037/0003-066X.38.4.379)
31. Azuma H. 1981 A note on cross-cultural study. *Qtrly. Newsl. Lab. Comp. Hum. Cogn*. **3**, 23–25.
32. Cole M. 2006 Internationalism in psychology: we need it now more than ever. *Am. Psychol*. **61**, 904–917. (doi:10.1037/0003-066X.61.8.904)
33. Palser ER, Lazerwitz M, Fotopoulou A. 2022 Gender and geographical disparity in editorial boards of journals in psychology and neuroscience. *Nat. Neurosci*. **25**, 272–279. (doi:10.1038/s41593-022-01012-w)
34. Roberts SO, Bareket-Shavit C, Dollins FA, Goldie PD, Mortenson E. 2020 Racial inequality in psychological research: trends of the past and recommendations for the future. *Perspect. Psychol. Sci*. **15**, 1295–1309. (doi:10.1177/1745691620927709)
35. Barker JS. 2016 Why 25 years? Notes on the long trajectory of Roger Barker's research in Oskaloosa. *Ecol. Psychol*. **28**, 39–55. (doi:10.1080/10407413.2016.1121744)
36. Barker, R. G. and Associates. 1978 *Environments, habitats, and human behavior: studies in ecological psychology and eco-behavioral science from the midwest psychological field station, 1947–1972*. San Francisco, CA: Jossey-Bass.
37. Barker RG. 1963 On the nature of the environment. *J. Soc. Issues.* **19**, 17–38. (doi:10.1111/j.1540-4560.1963.tb00456.x)
38. Barker RG, Wright HF. 1951 *One boy's day*. New York, NY: Harper & Brothers.
39. Barker RG. 1968 *Ecological psychology: concepts and methods for studying the environment of human behavior*. Stanford, CA: Stanford University Press.
40. Barker RG, Gump PV. 1964 *Big school, small school: high school size and student behavior*. Stanford, CA: Stanford University Press.
41. Perkins DV, Burns TF, Perry JC, Nielsen KP. 1988 Behavior setting theory and community psychology: an analysis and critique. *J. Community Psychol.* **16**, 355–372. (doi:10.1002/1520-6629(198810)16:4<355::AID-JCOP2290160402>3.0.CO;2-D)
42. Barker RG. 1989 Roger G. Barker. In *A history of psychology in autobiography* (ed. G Lindzey), pp. 3–35, vol. **8**. Stanford, CA: Stanford University Press.
43. Schoggen P. 1988 Commentary on Perkins, Burns, Perry, and Nielsen's "behavior setting theory and community psychology: an analysis and critique." *J. Community Psychol.* **16**, 373–386. (doi:10.1002/1520-6629(198810)16:4<373::AID-JCOP2290160403>3.0.CO;2-E)
44. Barker RG. 1979 Settings of a professional lifetime. *J. Pers. Soc. Psychol*. **37**, 2137–2157. (doi:10.1037//0022-3514.37.12.2137)
45. Hollan J, Hutchins E, Kirsh D. 2000 Distributed cognition: toward a new foundation for human-computer interaction research. *ACM Trans. Comput. Hum. Interact.* **7**, 174–196. (doi:10.1145/353485.353487)
46. Norman DA. 2013 *The design of everyday things: revised and expanded edition*. Cambridge, MA: MIT Press.
47. Boden MA. 2006 *Mind as machine: a history of cognitive science*. Oxford, UK: Oxford University Press.
48. Newell A. 1980 Physical symbol systems. *Cogn. Sci*. **4**, 135–183. (doi:10.1016/S0364-0213(80)80015-2)
49. Hutchins E. 2005 Material anchors for conceptual blends. *J. Pragmat*. **37**, 1555–1577. (doi:10.1016/j.pragma.2004.06.008)
50. Hutchins E. 1995 How a cockpit remembers its speeds. *Cogn. Sci*. **19**, 265–288. (doi:10.1016/0364-0213(95)90020-9)
51. Marr D. 1982 *Vision: a computational investigation into the human representation and processing of visual information*. San Francisco, CA: W. H. Freeman and Company.
52. Barker RG. 1965 Explorations in ecological psychology. *Am. Psychol.* **20**, 1–14. (doi:10.1037/h0021697)
53. Trasmundi SB, Baggs E, Toro J, Steffensen SV. 2024 Expertise in non-well-defined task domains: the case of reading. *Soc. Epistemol*. **38**, 13–27. (doi:10.1080/02691728.2023.2266690)
54. Chater N, Loewenstein G. 2023 The i-frame and the s-frame: how focusing on individual-level solutions has led behavioral public policy astray. *Behav. Brain Sci.* **46**. (doi:10.1017/S0140525X22002023)

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